

ADETORO DAMILOLA SIMBIAT

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PROFILE SUMMARY

AIOps Engineer with 5+ years experience spanning anomaly detection, automated incident response, backend systems, and applied machine learning. Master's in Engineering Artificial Intelligence at Carnegie Mellon University, with focus in ModelOps, automated monitoring, and AI-driven system reliability.

PROFESSIONAL EXPERIENCE

Algorisense — AI Engineer

Sept 2024 - Present

- Designed, developed, and optimized scalable backend components for fintech applications to keep the system responsive and reliable as user load and feature complexity grew, while integrating user-facing elements with production stability in mind
- Integrated LLM-based log anomaly detection into the production observability stack, leveraging semantic log understanding to surface irregular system behavior before user impact
- Built a proactive testing, deployment, and monitoring framework with automated error detection and routine security patching, catching issues before they reached users and reducing support ticket volume by 41.3% while maintaining error-free releases
- Fine-tuned and adapted pre-trained language models using LoRA and adapter-based tuning strategies to domain-specific fintech log data, reducing computational overhead while maintaining detection performance

AEIRG, Minna — AIOps Engineer

Oct 2022 - March 2024

- Built automated anomaly detection pipelines using Prometheus and Loki, transitioning incident response from reactive to proactive and reducing Mean Time to Resolution (MTTR) by 45% (from 4.2 to 2.3 hours)
- Applied AI-driven detection models to production log streams, flagging irregular system behavior ahead of failures and reducing unplanned downtime across critical services
- Designed and maintained real-time monitoring dashboards and alerting workflows surfacing performance deviations across distributed production services
- Evaluated and benchmarked log anomaly detection approaches including DeepLog, LogAnomaly, and BERT-based classifiers, informing model selection decisions for operational pipelines
- Collaborated cross-functionally with engineering teams to embed AI-based proactive detection into existing incident-response workflows, improving operational efficiency and alert signal quality

AI4CE, Minna — AI Research Assistant

May 2020- January 2021

- Built a real-time log collection pipeline to ingest sensor data from renewable energy systems, structuring and routing data for downstream anomaly analysis
- Developed an LSTM-based autoencoder anomaly detection model in TensorFlow/Keras, enabling early fault identification in renewable energy infrastructure with measurable reduction in undetected sensor failures
- Applied session-window log grouping and regex-based log normalization techniques to handle out-of-vocabulary (OOV) tokens, mirroring approaches later validated in production-grade systems like

LogLLM

- Evaluated model performance using F1-score, precision, and recall on imbalanced sensor datasets, documenting trade-offs between false positive rates and anomaly recall in high-stakes energy monitoring contexts

TECHNICAL SKILLS

- **Languages:** Python, C++, Bash/Linux
- **Frameworks & APIs:** FastAPI, Django, TensorFlow, Keras
- **Infrastructure & DevOps:** Docker, Kubernetes, Azure, CI/CD pipelines, automated security patching
- **Observability & AIOps:** Prometheus, Loki, real-time dashboards, alerting workflows, incident response automation, MTTR optimization
- **ML & Deep Learning:** LSTM autoencoders, anomaly detection, clustering, LLMs, BERT-based classifiers, RAG pipelines, LoRA fine-tuning
- **Databases & Caching:** Redis, vector databases (FAISS), SQL
- **Log Analysis:** Log parsing (Drain), session-window grouping, regex normalization, OOV handling, F1/precision/recall evaluation

PROJECTS

Failure Forecast: Predictive Failure Detection for High-Availability Infrastructure

A time series ML system that predicts infrastructure failure before it happens, using real SMART telemetry from 280K+ production hard drives (Backblaze Drive Stats). Combines XGBoost and LSTM models, cost based threshold selection, PSI drift monitoring, and a containerized FastAPI serving layer, evaluated on precision, recall, and PR AUC across multiple failure lead times. GitHub: [LINK](#)

Payment RCA Agent: Local LLM System for Autonomous Payment Failure Investigation

A locally hosted LLM agent that investigates failed payment transactions and generates structured root cause analysis reports, with zero external API dependency. Built with LangGraph for agentic orchestration, Ollama for local inference, and ChromaDB for retrieval over historical incidents, evaluated on classification precision, retrieval accuracy, and inference latency against a labeled synthetic test set. GitHub: [LINK](#)

EDUCATION

Carnegie Mellon University

August 2025 - September 2026

Masters in Engineering

Engineering Artificial Intelligence, Focus: MLOps, Natural Language Processing, AI for Optimization

Federal University of Technology, Minna

November 2018 - Sept 2024

Bachelor of Technology

Electrical Electronics Engineering , Focus: Anomaly Detection in Sensor Data

AWARDS

MINDS Scholarship, Dean's list, (Awarded to students with the top 90 percentile grades), Funmi Omo grant and mentorship program, PTDF Scholarship, Grace Hopper Conference Scholarship, UNIDO KPABEP Scholarship, Project support lead for ingressive for good(Futminna chapter)